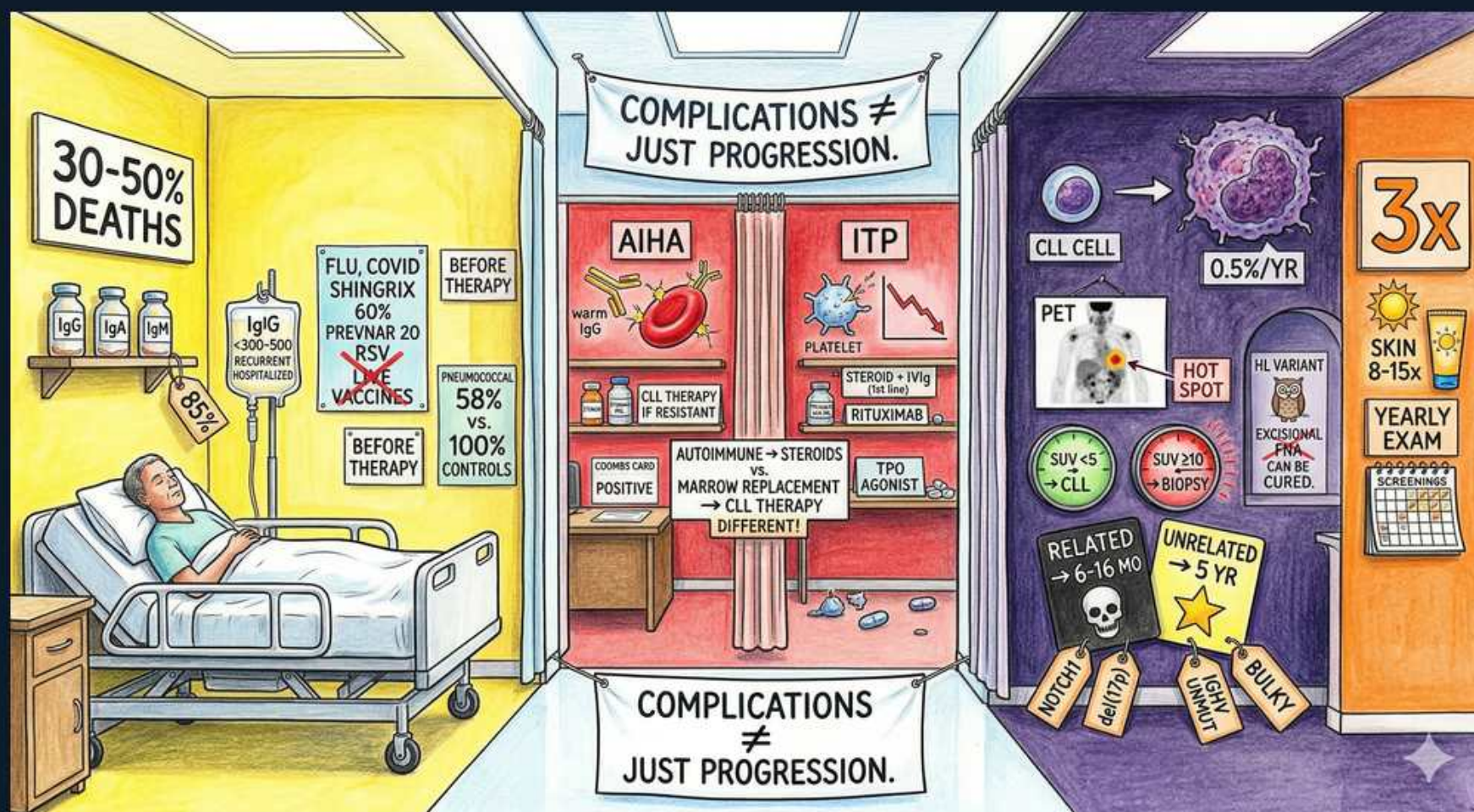


CLL — Infections, Autoimmune Cytopenias & Richter Transformation



1. Infection = Leading Cause of Death

WHAT YOU SEE

'30-50% DEATHS' sign, hospitalized patient in bed

WHAT IT ENCODES

Infection is the leading cause of death in CLL (~30-50%), driven by hypogammaglobulinemia and immune dysfunction. Both the disease and its therapies (BTK inhibitors, anti-CD20) increase infection risk.

BOARD TRAP

CLL patients die more often from infection than from the leukemia itself — immune failure, not tumor burden, is frequently the killer.

2. Hypogammaglobulinemia

WHAT YOU SEE

IgG/IgA low, '<300-500', recurrent infections, IVIG

WHAT IT ENCODES

Acquired hypogammaglobulinemia (low IgG) causes recurrent sinopulmonary infections with encapsulated organisms. IVIG replacement is indicated for IgG <300-500 plus recurrent serious/hospitalizing infections.

BOARD TRAP

Give IVIG for hypogammaglobulinemia WITH recurrent infections — not for a low IgG number alone. Replacement reduces bacterial infections, not viral.

3. Vaccinations

WHAT YOU SEE

FLU, COVID, SHINGRIX, RSV, PREVNAR 20, 'NO LIVE VACCINES', before therapy

WHAT IT ENCODES

Vaccinate early (ideally BEFORE therapy/anti-CD20) — pneumococcal (PCV20), influenza, COVID, recombinant zoster (Shingrix), RSV. Avoid LIVE vaccines. Anti-CD20 blunts vaccine responses for months.

BOARD TRAP

Live vaccines (e.g., live zoster, yellow fever) are contraindicated — use recombinant Shingrix. Vaccinate BEFORE rituximab/obinutuzumab, which abolish humoral response.

4. Vaccine Response Timing

WHAT YOU SEE

'58% vs 100% controls', before therapy comparison

WHAT IT ENCODES

Vaccine immunogenicity is markedly reduced in CLL (e.g., ~58% response vs ~100% in controls) and is nearly abolished after anti-CD20 therapy — reinforcing the rule to immunize before treatment.

BOARD TRAP

Vaccinating during/after anti-CD20 wastes the opportunity — B-cell depletion prevents antibody formation for 6+ months. Timing is the board point.

5. AIHA (Autoimmune Hemolysis)

WHAT YOU SEE

'AIHA' panel, red cells, steroids+IVIG+rituximab, Coombs+

WHAT IT ENCODES

Warm autoimmune hemolytic anemia is the most common CLL autoimmune complication — Coombs (DAT) positive, spherocytes, high LDH/retic, low haptoglobin. Treat with steroids ± rituximab/IVIG; not a marrow-failure cause.

BOARD TRAP

AIHA anemia is IMMUNE destruction, not marrow infiltration — it does NOT upstage to Rai III. Fludarabine can trigger/worsen AIHA — a key drug caution.

6. ITP (Immune Low Platelets)

WHAT YOU SEE

'ITP' panel, platelet, steroid+IVIG+TPO agonist, rituximab

WHAT IT ENCODES

Immune thrombocytopenia in CLL is antibody-mediated platelet destruction with normal/increased marrow megakaryocytes. Treat with steroids, IVIG, rituximab, or thrombopoietin receptor agonists.

BOARD TRAP

Immune low platelets does NOT mean Rai stage IV — that requires marrow-failure thrombocytopenia. Distinguishing immune vs infiltrative cytopenia changes staging and therapy.

7. Complications ≠ Progression

WHAT YOU SEE

Banner top and bottom: 'COMPLICATIONS ≠ JUST PROGRESSION'

WHAT IT ENCODES

Autoimmune cytopenias are immune-mediated complications distinct from disease progression/marrow replacement. They are treated with immunosuppression (steroids/rituximab), not necessarily by escalating CLL-directed therapy.

BOARD TRAP

Treating immune cytopenia by intensifying CLL chemo is wrong — immunosuppression addresses the autoimmune mechanism. Marrow-replacement cytopenia, by contrast, calls for CLL therapy.

8. Richter Transformation

WHAT YOU SEE

CLL cell → large transformed cell, '0.5%/YR', PET hot spot

WHAT IT ENCODES

Richter transformation = CLL evolving into aggressive diffuse large B-cell lymphoma (rarely Hodgkin). Incidence ~0.5%/yr. Presents with rapid nodal growth, B symptoms, soaring LDH. PET-directed biopsy of the SUV-high node confirms.

BOARD TRAP

A single rapidly enlarging node with very high SUV on PET signals Richter — biopsy that node. NOTCH1/TP53 and prior chemo increase risk; prognosis is poor.

9. PET-Directed Biopsy (SUV)

WHAT YOU SEE

'SUV>5 = CLL' vs 'SUV>10 = BIOPSY', hot spot inset

WHAT IT ENCODES

PET helps target Richter biopsy — SUV ≥ 10 strongly suggests transformation and directs excisional biopsy of the hottest node. Untransformed CLL nodes have low-to-moderate uptake.

BOARD TRAP

Don't biopsy a random or convenient node — PET points to the highest-SUV site, where transformation is most likely to be captured. Excisional > core biopsy.

10. Clonal Relationship Prognosis

WHAT YOU SEE

'RELATED 6-16 MO' vs 'UNRELATED 5 YR', clonal tags

WHAT IT ENCODES

Richter DLBCL clonally RELATED to the original CLL has dismal prognosis (median ~6-16 months); clonally UNRELATED (de novo) DLBCL behaves like ordinary DLBCL with ~5-year survival potential.

BOARD TRAP

Not all Richter is uniformly fatal — clonally unrelated transformation can be cured like standard DLBCL. Clonal relationship (IGHV sequencing) is the prognostic discriminator.

11. Secondary Skin Cancers

WHAT YOU SEE

'3x' sign, sun/SPF, 'SKIN 8-15x', yearly exam

WHAT IT ENCODES

CLL's immune dysfunction raises secondary cancer risk — notably non-melanoma skin cancers (up to 8-15x) and melanoma. Recommend sun protection and yearly dermatologic surveillance.

BOARD TRAP

Aggressive skin cancers (squamous cell, Merkel) behave worse in immunosuppressed CLL patients — don't dismiss skin lesions; surveillance and early excision matter.

12. Encapsulated Organism Risk

WHAT YOU SEE

Pneumococcal vaccine emphasis, hypogammaglobulinemia panel

WHAT IT ENCODES

Hypogammaglobulinemia predisposes to encapsulated bacteria (*Streptococcus pneumoniae*, *Haemophilus*, *Neisseria*) — hence pneumococcal (PCV20) vaccination and prompt treatment of respiratory infections.

BOARD TRAP

The encapsulated-organism vulnerability mirrors asplenia/complement deficiency — antibody, not neutrophil, defense is the gap. PCP/antiviral prophylaxis tracks with specific therapies (e.g., purine analogs).
